Mzuzu University

Department of Mathematics and Statistics

MATH 1101 - Precalculus

End of Semester Examination

Monday, 8th August 2022 Time Allowed: 3 hours (08:00 - 11:00 hrs)

INSTRUCTIONS

1. Write your registration number, programme and the name of lecturer in the spaces provided below.
2. Registration number should also appear on top of every page.
3. Attempt **all** questions.
4. Calculators are **not** allowed.
5. Check that the paper has **8** questions.
6. Show **all** your working and use appropriate notation.

Registration Number: _____

Programme: : _____

Lecturer's Name: : _____

Repeater: : Yes ☐ No ☐

Question:	1	2	3	4	5	6	7	8	Total
Points:	12	14	12	10	7	10	12	7	84
Score:									

QUESTIONS

1. (a) Factorise $x^3a^2 - 8y^3a^2 - 4x^3b^2 + 32y^3b^2$ completely. $\boxed{4}$

(b) Rewrite the expression using rational exponents: $\sqrt[3]{\sqrt{x}} + \sqrt{\sqrt[3]{x}}$. $\boxed{3}$

(c) Rationalize the denominator and simplify: $\boxed{5}$

$$\frac{ax^2 + bx - axy - by}{\sqrt[3]{x^2} + \sqrt[3]{xy} + \sqrt[3]{y^2}}.$$

2. Determine all of the real-number solutions for each of the following equations:

(a) $3x^{\frac{4}{3}} + 4x^{\frac{2}{3}} - 4 = 0,$ $\boxed{6}$

(b) $\sqrt{2y+5} - \sqrt{8y+25} + \sqrt{2y+8} = 0.$ $\boxed{8}$

3. Solve the following inequalities and specify the answer using interval notation:

(a) $(1-x)^2 < \frac{9}{25}$; $\boxed{6}$

(b) $\left|\frac{5x-8}{4} - \frac{2x-7}{2}\right| \geq 3$. $\boxed{6}$

4. (a) Determine if the graph of $|y| = 2x - 4$ is symmetric about

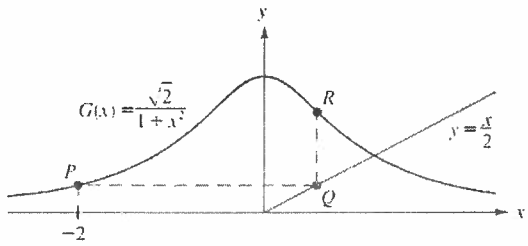
   i. $x-$axis, $\boxed{2}$

   ii. $y-$axis, $\boxed{2}$

   iii. origin. $\boxed{2}$

   (b) Suppose that (a, b) is a point on the graph of $y = f(x)$, specify the corresponding point on the graph of $y = f(3 - x) + 1$. $\boxed{4}$

5. Given that PQ is parallel to x-axis and RQ is parallel to y-axis, determine the coordinates of the points P, Q and R in the figure below. $\boxed{7}$



6. (a) Let $f(x) = ax^2 + bx + c$. Find the difference quotient $\dfrac{f(x+h) - f(x)}{h}$, hence evaluate it when $h = 0$. $\boxed{5}$

(b) Functions f and g are given by $f(x) = \sqrt{x+2}$ and $g(x) = \ln(1 - x^2)$. Find the function $(g \circ f)(x)$ and describe its domain. $\boxed{5}$

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7. Graph the following functions and specify their range, intercept(s), and asymptote(s):

(a) $f(x) = -x^2 + 4x - 3$, $\boxed{6}$

(b) $f(x) = \begin{cases} 2^{-x} & \text{if } x \leq -2 \\ -\ln(x+2) & \text{if } x > -2. \end{cases}$ 6

8. (a) Solve for x: $\ln 4e^{3x} + \ln(e^{3x} - 4) = \ln 9$.

(b) Solve the inequality: $\log_2 x + \log_2(x+1) - \log_2(2x+6) < 0$. $\boxed{7}$

END OF QUESTIONS